MAT2540, Classwork6, Spring2026

6.1 The Basics of Counting (Skip trees)

1. The **Product Rule** With m Tasks

   Suppose that a procedure can be broken down into a sequence of m tasks $T_1, T_2, \cdots, T_m$. If each task T_i can be done in n_i ways, for $i = 1,2,\cdots m$, then there are $n_1 \cdot n_2 \cdot n_3 \cdot \cdots \cdot n_m$ ways to do the procedure.

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2. How many different license plates can be made if each plate contains a sequence of three uppercase English letters followed by three digits (and no sequences of letters are prohibited, even if they are obscene)?

   26 10

   $26 \cdot 26 \cdot 26 \cdot 10 \cdot 10 \cdot 10 = 17576000$

   $f(x_1) = y_{100}$, $f(x_3) = y_{100}$, $x_1, x_3 \to y_{100}$

   $x_3 \to y_1$, $x_5 \to y_2$, $x_1 \to y_3$, $x_2 \to y_4$, $\cdots$, $x_{100} \to y_n$

3. Example: Counting One-to-One Functions. $f(x) = y$, if f is 1-1, same output gets same input

   How many one-to-one functions are there from a set with m elements (domain) to one with n elements (codomain)?

   If $m > n$, there are no one-to-one function from a set of m to a set of n elements.

   If $m \le n$, m objects

   $y_1 \quad y_2 \quad y_3 \quad \cdots \quad y_n$:

   1st object: n ways $\cdot$ 2nd object: $n-1$ ways $\cdots$ mth object: $n-m+1$ ways

   We have $n \cdot (n-1) \cdot (n-2) \cdot (n-m+1)$ ways of one-to-one function

4. Given a set S. Then the number of the elements in S is denoted by $|S|$.

   $a \in \mathbb{R}^2$ or $\mathbb{R} \times \mathbb{R}$

   $a(x_1, y_1)$: $\mathbb{R} \to \mathbb{R}$

5. The product rule in terms of sets: If $A_1, A_2, \cdots, A_m$ are finite sets, then the number of elements in the Cartesian product of these sets $A_1 \times A_2 \times \cdots \times A_m$ is the product of the number of elements in each set: $|A_1 \times A_2 \times A_3 \times \cdots \times A_m| = |A_1| \cdot |A_2| \cdot |A_3| \cdot \cdots \cdot |A_m|$

6. Example: Counting Subsets of a Finite Set. $S = \{1,2,3\}$, $|S| = 3$, $2^3 = 8$

   $\{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}, \{\phi\}$

   Use the product rule to show that the number of different subsets of a finite set S is $2^{|S|}$.

   Let $|S| = n$ be a finite number

   $S = \{a_1, a_2, a_3, a_4, \cdots, a_n\}$, for each element of S, either you pick it or you don't

   the ways to pick: $2 \cdot 2 \cdot 2 \cdot 2 \cdots \cdot 2 = 2^n = 2^{|S|}$

7. Please read the description about DNA and Genomes in Example 11 on p.409 before doing HW5.

8. The **Sum Rule** With m Tasks

   Suppose that a task can be done in one of n_1 ways, in one of n_2 ways,... , or in one of n_m ways, where none of the set of n_i ways of doing the task is the same as any of the set of n_j way, for all pairs i and j with $1 \le i < j \le m$. Then the number of ways to do the task is $n_1 + n_2 + n_3 + \cdots + n_m$.

9. The sum rule in terms of sets: If $A_1, A_2, \cdots, A_m$ are pairwise disjoint finite sets, then the number of elements in the union of these sets $A_1 \cup A_2 \cup A_3 \cup \cdots \cup A_m$ is the sum of the numbers of elements in the sets:

   $|A_1 \cup A_2 \cup A_3 \cup \cdots \cup A_m| = |A_1| + |A_2| + |A_3| + \cdots + |A_m|$

10. Let $A_1 = \{1, 2, 3\}, A_2 = \{4, 5, 6, 9\}$, and $A_3 = \{7, 8\}$. Find $|A_1| =$ 3, $|A_2| =$ 4, $|A_3| =$ 2,

$A_1 \cup A_2 \cup A_3 =$ $\{1, 2, 3, 4, 5, 6, 9, 7, 8\}$ and $|A_1 \cup A_2 \cup A_3| =$ 9.

$|A_1| + |A_2| + |A_3| = 3 + 4 + 2 = 9$

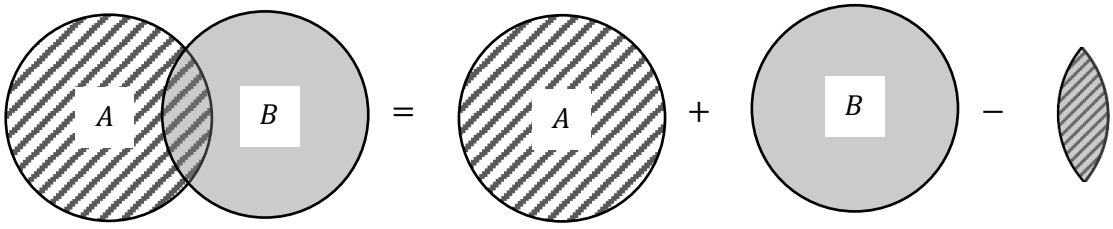
11. The **Subtraction Rule**

If a task can be done in either n_1 ways or n_2 ways, then the number of ways to do the task is _____

_____

12. The **Principle of Inclusion-Exclusion** for two sets:

Let $\boldsymbol{A}$ and $\boldsymbol{B}$ be sets. We have $|A __ B| = |A| + |B| - |A __ B|$



13. Let $A = \{1, 2, 3\}$, $B = \{1, 3, 5\}$. Show that $|A \cup B| = |A| + |B| - |A \cap B|$.

14. The Principle of Inclusion-Exclusion for **three** sets:

Let $\boldsymbol{A}$, $\boldsymbol{B}$, and $\boldsymbol{C}$ be sets. Then we have

$$|A \cup B \cup C| = |A| + |B| + |C| - |__ \cap __| - |__ \cap __| - |__ \cap __| + |__ \cap __ \cap __|.$$

15. The **Division Rule**

There are $\frac{n}{d}$ ways to do a task if it can be done using a procedure that can be carried out in n ways, and for every way w, exactly d of the n ways correspond to way w.

16. How many different ways are there to seat four people around a circular table, where two seatings are considered the same when each person has the same left neighbor and the same right neighbor?